# 崑山科技大學電機工程系大學部四年制國際產學專班課程科目表

課程名稱（講授時數-實習時數-學分數）

114.04.17系課程委員會通過
114.04.22院課程委員會通過
114.07.22校課程委員會通過
110學年度入學適用

| 1st Academic Year | | 2nd Academic Year | | 3rd Academic Year | | 4th Academic Year | |
|---|---|---|---|---|---|---|---|
| 1st Semester | 2nd Semester | 1st Semester | 2nd Semester | 1st Semester | 2nd Semester | 1st Semester | 2nd Semester |
| **General Compulsory Courses 共同必修(Total: 25 Cs)** | | | | | | | |
| 實用生活華語(一)<br>Situational Chinese (I)<br>(6-0-3)<br><br>職場溝通華語(一)<br>Practical Workplace Chinese(I)<br>(4-0-2)<br><br>職場倫理<br>Workplace Ethics<br>(2-0-2) | 實用生活華語(二)-進階<br>Situational Chinese (II)-Advanced Level<br>(3-0-3)<br><br>職場溝通華語(二)<br>Workplace Chinese(II)<br>(3-0-3)<br><br>台灣歷史與文化<br>History and Culture of Taiwan<br>(3-0-3) | 華語口語與表達<br>Oral Performance in Mandarin Chinese (3-0-3) | 專業華語<br>Professional Chinese about EE<br>(3-0-3) | 高級華語聽力閱讀<br>Advanced Mandarin Listening and Reading<br>(3-0-3) | | | |
| 12-0-7 | 9-0-9 | 3-0-3 | 3-0-3 | 3-0-3 | 0-0-0 | 0-0-0 | 0-0-0 |
| **Professional Compulsory Courses 專業必修(Total: 53Cs)** | | | | | | | |
| 普通物理與實驗<br>Physics and Experiment<br>(2-2-4)<br><br>基本電學及實習<br>Basic Circuits and Experiment<br>(2-2-4) | 電子學及實習(一)<br>Electronics and Laboratory (1)<br>(2-2-4)<br><br>電路學與實習<br>Circuit and the Experiment<br>(2-2-4) | 電子學及實習(二)<br>Electronics and Laboratory (1I)<br>(2-2-4)<br><br>職場實習一<br>Field Practice (I)<br>(0-24-6) | 材料科學導論<br>Fundamentals of Materials Science and Engineering<br>(3-0-3)<br><br>自動化概論<br>Introduction to Automation<br>(3-0-3)<br><br>職場實習二<br>Field Practice (II)<br>(0-24-6) | 工程材料學<br>Engineering Materials<br>(3-0-3)<br><br>半導體製程及實習<br>Semiconductor Technology and Practice<br>(1-2-3)<br><br>職場實習三<br>Field Practice (III)<br>(0-24-6) | 半導體元件<br>Semiconductor Device<br>(3-0-3) | | |
| 4-4-8 | 4-4-8 | 2-26-10 | 6-24-12 | 4-26-12 | 3-0-3 | 0-0-0 | 0-0-0 |
| **Selective Courses 選修(Total: 50Cs)** | | | | | | | |
| | 工業安全與衛生<br>Industrial Safety and Health<br>(3-0-3) | 基礎量測及實習<br>The Fundamental of Measurement | | | 專題實作(一)<br>Project (I)<br>(2-2-4)<br><br>電力電子學 | 專題實作(二)<br>Project (II)<br>(2-2-4)<br><br>LED 原理及 | 感測元件與應用<br>Sensors and Applications<br>(3-0-3) |

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|---|---|---|---|---|---|---|---|
| | | and Experiments (1-2-3) | | | 實習 Experiments of Power Electronics (1-2-3)<br><br>職場實習四 Field Practice (IV) (0-24-6) | 應用 Principles and Applications of LED (3-0-3)<br><br>高亮度 LED 技術及實習 High-Brightness LED Technology and Practice (1-2-3)<br><br>職場實習五 Field Practice (V) (0-24-6) | 表面工程技術 Surface Engineering Technology (3-0-3)<br><br>計算機概論 Introduction to Computer (1-2-3)<br><br>職場實習六 Field Practice (VI) (0-24-6) |
| 0-0-0 | 3-0-3 | 1-2-3 | 0-0-0 | 0-0-0 | 3-28-13 | 6-28-16 | 7-26-15 |
| Minimum for Total Credits: 128 Cs | | | | | | | |

Note 備註:

- (Le-Lb-Cs), Le: Lecture hours, Lb: Lab hours, Cs: Credits。
  課程名稱（講授時數-實習時數-學分數）
- The university reserves the right to make changes to the courses offered without prior notice.
  本校保留課程調整之權利。
- This class is a special class, so the courses offered are exactly 128 credits. If students wish to take elective courses in other classes of the university, they can apply for other courses after they are approved by the class committee. The total credits of the courses that students can take can exceed 128 credits.
  本班屬於專班性質，因此所開設的課程剛好為128 學分，如有學生希望能修習本校其他班級之選修課程時，得提出申請，並經專班委員會通過後，即可選修其他課程，不會僅限128 學分。
- If a student is unable to continue the internship due to reasons, and cannot transition to another company for internship, or if the student is not suitable for an internship in the current semester, they may choose to take the following courses as a substitute.
  若學生因故中止實習且無法銜接至其他公司進行實習，或學生該學期不適合實習時，得選修下列課程替代之。

| 編號 | 課程名稱 | 學分數 |
|---|---|---|
| 1 | 數位電子乙級技術士學科 | (3-0-3) |
| 2 | 數位電子乙級技術士術科 | (1-2-3) |
| 3 | 電力電子乙級技術士學科 | (3-0-3) |
| 4 | 電力電子乙級技術士術科 | (1-2-3) |
| 5 | 真空半導體設備人機介面及實習 | (1-2-3) |
| 6 | 綠能科技與實作 | (2-1-3) |
| 7 | 綠色能源與產業商機 | (3-0-3) |
| 8 | 機電整合實務 | (3-0-3) |
| 9 | 可程式控制器進階控制實習 | (0-3-3) |
| 10 | 電腦硬體裝修檢定學術科 | (3-0-3) |

| 編號 | 課程名稱 | 學分數 |
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| 11 | 影像視覺檢測 | (3-0-3) |
| 12 | 電機機械實習 | (0-2-2) |
| 13 | VB 監控實習 | (3-0-3) |
| 14 | 真空技術士認證輔導 | (3-0-3) |
| 15 | 電力系統模擬 | (3-0-3) |
| 16 | LED 照明專論 | (3-0-3) |
| 17 | 電力品質分析與模擬 | (3-0-3) |
| 18 | 數位影像處理 | (3-0-3) |
| 19 | 半導體製程 | (3-0-3) |
| 20 | AI 機電視覺分析 | (3-0-3) |
| 21 | 電力暫態分析與模擬 | (3-0-3) |